

Cryptography Assignment - 1

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Section:- 4B

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Q.1 ⇒ Assets

Confidentiality

Integrity

Availability

Products

No

Yes

Yes

Agreements
with Suppliers

Yes

Yes

No

Client Invoices

Yes

Yes

No

Cash Registers

No

Yes

Yes

Q.2 ⇒ Assets

Confidentiality

Integrity

Availability

Medical file

Yes

Yes

Yes

University website

No

Yes

Yes

Aircraft design dev.

Yes

Yes

No

Home Camera

Yes

Yes

Yes

City Surveillance Camera

No

Yes

Yes

SKI broadcast station

No

No

Yes

Q.3 ⇒ (i) Student steal exam paper → Disclosure
Confidentiality attack)
(ii) check of Rs.10 changed to Rs.100 → Modification
(Integrity attack)

(iii) Hundreds of Spammails $\rightarrow$ Availability attack (DDoS)

Q.4 $\rightarrow$ (i) Login with ID card & password $\rightarrow$ authentication

(ii) Auto Logout after 2 hours $\rightarrow$ access control

(iii) Professor checks ID before sending grades $\rightarrow$ authentication

Q.5 $\rightarrow$ Threat $\rightarrow$ The company can collect and combine personal data from all magazines $\rightarrow$ privacy breach.

* Prevention $\rightarrow$ Share minimal personal data, or use separate email for each subscription.

Q.6 $\rightarrow$ $3 \xrightarrow{e} 22$

$1 \xrightarrow{e} 1$

$\Rightarrow$ Using equations from Affine cipher Algorithm:

$$22 = (3K_1 + K_2) \bmod 31 \quad - (i)$$

$$1 = (1K_1 + K_2) \bmod 31 \quad - (ii)$$

$\Rightarrow$ Solving both equations: - Sub. (ii) from (i):

$$21 = (2K_1) \bmod 31$$

$$\therefore 2K_1 = 52$$

$$\boxed{K_1 = 26}$$

Now, for K_2 : $22 = (3K_1 + K_2) \bmod 31$

$$22 = (78 + K_2) \bmod 31$$

$$\Rightarrow 78 + K_2 = 84$$

$$\boxed{K_2 = 6}$$

Q.7 → Cipher

Caesar cipher

Pairs Needed

1

Reason:

Only one shift value.

Affine cipher

2

Need two eqns to find a & b.

Vigenere cipher

1

Need one pair Key letter.

Q.8 → Plaintext :- life is full of surprises

Keyword :- health [7, 4, 0, 11, 19, 7]

Plain Text	l	i	f	e	i	s	f	u	l	l	o	f	s	u	r	p	r	i	s	e	s
P's values	11	8	5	4	8	18	5	20	11	11	14	5	18	20	17	15	17	8	18	4	18
Key value	7	4	0	11	19	7	7	4	0	11	19	7	7	4	0	11	19	7	7	4	0
C's value	18	12	5	15	1	25	12	24	11	22	7	12	25	24	17	0	10	15	25	8	18
Cipher text	S	M	F	P	B	Z	M	Y	L	W	H	M	Z	Y	R	A	K	P	Z	I	S

Q.9 → Plaintext ⇒ Cryptography is fun.

for $Z_{26}^* = \{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25\}$

Plain Text	P's value	$(i \bmod 12)$	$(i \bmod 26)$	Mult Key (a_i)	Additive Key (b_i)	Cipher Text
C	2	0	0	1	0	C
R	17	1	1	3	1	A
Y	24	2	2	5	2	S
P	15	3	3	7	3	E
T	19	4	4	9	4	T
O	14	5	5	11	5	D
G	6	6	6	15	6	S

R	17	7	7	17	7	K
A	0	8	8	19	8	I
P	15	9	9	21	9	M
H	7	10	10	23	10	P
Y	24	11	11	25	11	N
I	8	0	12	1	12	U
S	18	1	13	3	13	P
F	5	2	14	5	14	N
V	20	3	15	7	15	Z
N	13	4	16	9	16	D

Q.10 $\Rightarrow$ In Hill cipher; $\boxed{\text{Ciphertext} = \text{Plaintext} \times \text{Key}}$

* If plaintext = Identity matrix

* the ciphertext = $C = K \times I = K$

* Chosen Plaintext Attack $\Rightarrow$ By sending I as plaintext, attacker directly gets the key matrix K; breaking the cipher easily.